A04 Analysis of the movie “Arrival” through the lens of NLP

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Introduction

When it comes to language, I think most of us believe that we can just translate word-for-word when trying to understand a different language. I know for a fact that there are words that are impossible to translate from one language to another, and we also need to consider non-verbal languages such as sign language. In the movie Arrival, we can appreciate how complex communication is between humans and the beings from a different planet. The film follows linguist Dr. Louise Banks and her attempts to communicate with the extraterrestrials, but we can see a glimpse of the different approaches of the different nations dealing with the same issue. Of course, to add some drama into the movie, cooperations between nations come to a halt, and everyone starts working by themselves.

The challenges that we can observe in the movie that are similar to the ones we are presented with in Natural Language Processing include: ambiguity, cultural interpretation, symbol grounding, and structural differences. The extraterrestrials' language looks complex, and Dr. Louise uses simple words to make sense of what they are trying to communicate. There is progress with her approach, but like in real life, there are also limitations.

NLP Challenges in the film

In Arrival, humanity is dealing with a crisis of global panic because twelve extraterrestrial spacecraft appear around the world. Every government and nations are trying to understand the intentions of the visitors. When Dr. Louise Banks is recruited to decode their language using written language, she has to overcome several challenges. The similar challenges encountered in the movie and NLP include:

Ambiguity and context

One of the main challenges in the movie and NLP is ambiguity. This issue arises when we have words that have multiple meanings, and if we don't have context, we can misinterpret the actual meaning of the word. In the movie, the word "weapon" is the main example of this challenge. Dr. Louise tries to explain to her superiors that the extraterrestrials are probably talking about a tool and not a weapon, but without context, the whole situation escalates. In the English language, examples of this type of word include bank (financial institution or the side of the river) and date (fruit, calendar day, or romantic outing). NLP models can use contextual embeddings, semantic analysis, and syntactic & pragmatic analysis to help resolve this issue. But as we see in the movie, a small misinterpretation can cause a huge issue

Cultural and conceptual differences

As every culture in the world, the extraterrestrials have their own concept of reality and time. They experience time in a non-linear way. I was a little bit confused about this after I watched the whole plot envelop, but if every culture in the world is different, it makes sense that these beings and their reality are completely different from ours. When it comes to NLP, the cultural and conceptual variations present a challenge. Aspects like sentiment, sarcasm, and idiomatic expression vary from culture to culture. There are expressions that might have a negative connotation in one place, but not in another. Systems that are trained on one dataset may struggle with different linguistic norms or inherit biases that will complicate interpretation.

Symbol grounding problem

In the movie, Dr. Louise tries to use the word "human" and make the extraterrestrials understand that's what they are, but has difficulty connecting the symbols to real-world meaning. Her solution to make them understand "human", "Louise", and "Ian" was to take off her protective suit and show them her face. She was able to accomplish a better understanding after this, but this issue is called the symbol grounding problem. We learn how to connect symbols (words) to real-world objects as we grow. If I ask my three-year-old cousin to show me where her cat is, she will make that connection between the word and the pet. NLP systems still lack in this aspect because they cannot make the connection that requires direct experience.

Nonlinear structure and representation

The human language is processed sequentially; we put one word after another one until we manage to communicate with another individual. The extraterrestrials don't use a sequential process; they use circular forms that convey entire ideas at once. This way of communication is similar to sentence embeddings in deep learning because meaning is compressed into a vector representation. Word embedding allows models to represent relationships mathematically, so it is similar to the circular forms because they are compact but full of information.

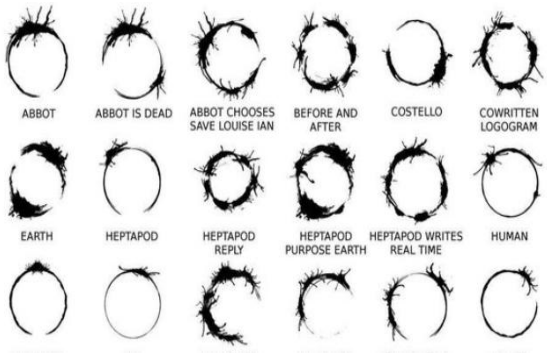
Communication methods in the movie

When Dr. Louise first listens to the extraterrestrials' recording from Colonel Weber, she determines that audio wasn't going to be enough to decipher their language. On the first encounter she had with the extraterrestrials, she wasn't able to do much because of the shock of looking at something that is not from our planet, but on the second encounter she decided to use a writing board. The approach Dr. Louise is trying to apply is similar to rule-based NLP systems, this type of systems rely on predefined grammatical rules and lexicons.

Dr. Louise starts collecting more data and finding patterns. This stage is similar to statistical NLP because meaning is inferred from frequency and probability distributions. We can observe how Dr. Louise starts to understand the language more and more, not by memorizing but by understanding the structure. This type of progress is similar to the evolution of deep NLP because neural networks learn complex representations from large datasets. In the movie, we can appreciate how language is transformative and not mechanical; it is similar to NLP systems because they simulate understanding through pattern recognition.

The movie Arrival shows us some of the real-world challenges in Natural Language Processing. The progression of communications in the movie is similar to the development of NLPs, and their challenges are something these systems struggle with in real life. In the movie, we understand that language is complex and is a reflection of perception, cognition, and shared experience.

Images from the movie



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